

Student Profile using JavaScript

Aim: To create a JavaScript Student class and dynamically display student details on a webpage using DOM manipulation and event handling.

Description: The application accepts Name, Roll Number, Department and CGPA from the user. A Student object is created from the supplied values. When the button is clicked, JavaScript dynamically creates HTML elements and displays the student profile.

Requirements: HTML, CSS, JavaScript and a web browser.

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Profile</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="container">
    <h1>Student Profile</h1>
    <div class="form-box">
      <input type="text" id="name" placeholder="Enter Name">
      <input type="text" id="rollNo" placeholder="Enter Roll Number">
      <input type="text" id="department" placeholder="Enter Department">
      <input type="number" id="cgpa" step="0.01" placeholder="Enter CGPA">
      <button id="createBtn">Create Student Profile</button>
    </div>
    <div id="profileContainer"></div>
  </div>
  <script src="script.js"></script>
</body>
</html>
```

CSS

```
* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}

body {
  font-family: Arial, sans-serif;
  background: #f2f4f7;
  min-height: 100vh;
  display: flex;
  justify-content: center;
  align-items: center;
}

.container {
  width: 90%;
  max-width: 650px;
  text-align: center;
}

h1 {
  margin-bottom: 20px;
}

.form-box {
  background: white;
  padding: 25px;
  border-radius: 12px;
  box-shadow: 0 4px 15px rgba(0, 0, 0, 0.12);
}

input {
  width: 100%;
```

```

padding: 12px;
margin: 7px 0;
border: 1px solid #ccc;
border-radius: 6px;
font-size: 16px;
}

button {
margin-top: 12px;
padding: 12px 20px;
border: none;
border-radius: 6px;
background: #2563eb;
color: white;
font-size: 16px;
cursor: pointer;
}

button:hover {
background: #1d4ed8;
}

.student-card {
margin-top: 25px;
background: white;
padding: 25px;
border-radius: 12px;
box-shadow: 0 4px 15px rgba(0, 0, 0, 0.12);
text-align: left;
}

.student-card h2 {
text-align: center;
margin-bottom: 20px;
}

.student-row {
display: flex;
padding: 10px 0;
border-bottom: 1px solid #eee;
}

.student-row .label {
width: 35%;
font-weight: bold;
}

.student-row .value {
width: 65%;
}

```

JavaScript

```

class Student {
  constructor(name, rollNumber, department, cgpa) {
    this.name = name;
    this.rollNumber = rollNumber;
    this.department = department;
    this.cgpa = cgpa;
  }
}

document.getElementById("createBtn").addEventListener("click", function () {
  const name = document.getElementById("name").value.trim();
  const rollNumber = document.getElementById("rollNo").value.trim();
  const department = document.getElementById("department").value.trim();
  const cgpa = document.getElementById("cgpa").value.trim();

  if (!name || !rollNumber || !department || !cgpa) {
    alert("Please enter all student details.");
    return;
  }

  const student = new Student(name, rollNumber, department, cgpa);

  const profileContainer = document.getElementById("profileContainer");
  profileContainer.innerHTML = "";

  const card = document.createElement("div");
  card.className = "student-card";

  const title = document.createElement("h2");
  title.textContent = "Student Profile";

```

```

card.appendChild(title);

const details = [
  ["Name", student.name],
  ["Roll No", student.rollNumber],
  ["Department", student.department],
  ["CGPA", student.cgpa]
];

details.forEach(function ([label, value]) {
  const row = document.createElement("div");
  row.className = "student-row";

  const labelElement = document.createElement("div");
  labelElement.className = "label";
  labelElement.textContent = label + " :";

  const valueElement = document.createElement("div");
  valueElement.className = "value";
  valueElement.textContent = value;

  row.appendChild(labelElement);
  row.appendChild(valueElement);
  card.appendChild(row);
});

profileContainer.appendChild(card);
});

```

Expected Output

Field	Value
Name	Rapeti.Srivalli
Roll No	A24126552108
Department	CSE (AI & ML)
CGPA	8.42

Learning Outcomes: JavaScript classes and objects, DOM selection, dynamic HTML element creation, modifying element content, adding elements to the webpage, and event handling.